بسم الله الرحمن الرحيم

Spleen

Introduction

- The spleen is the largest single lymphoid tissue in the body and the only one involved in filtration of blood making it an important organ in defense against blood-borne antigens.
- Located in the left upper abdominal quadrant or left hypochondrium of the abdominal cavity at the level of 11th rib

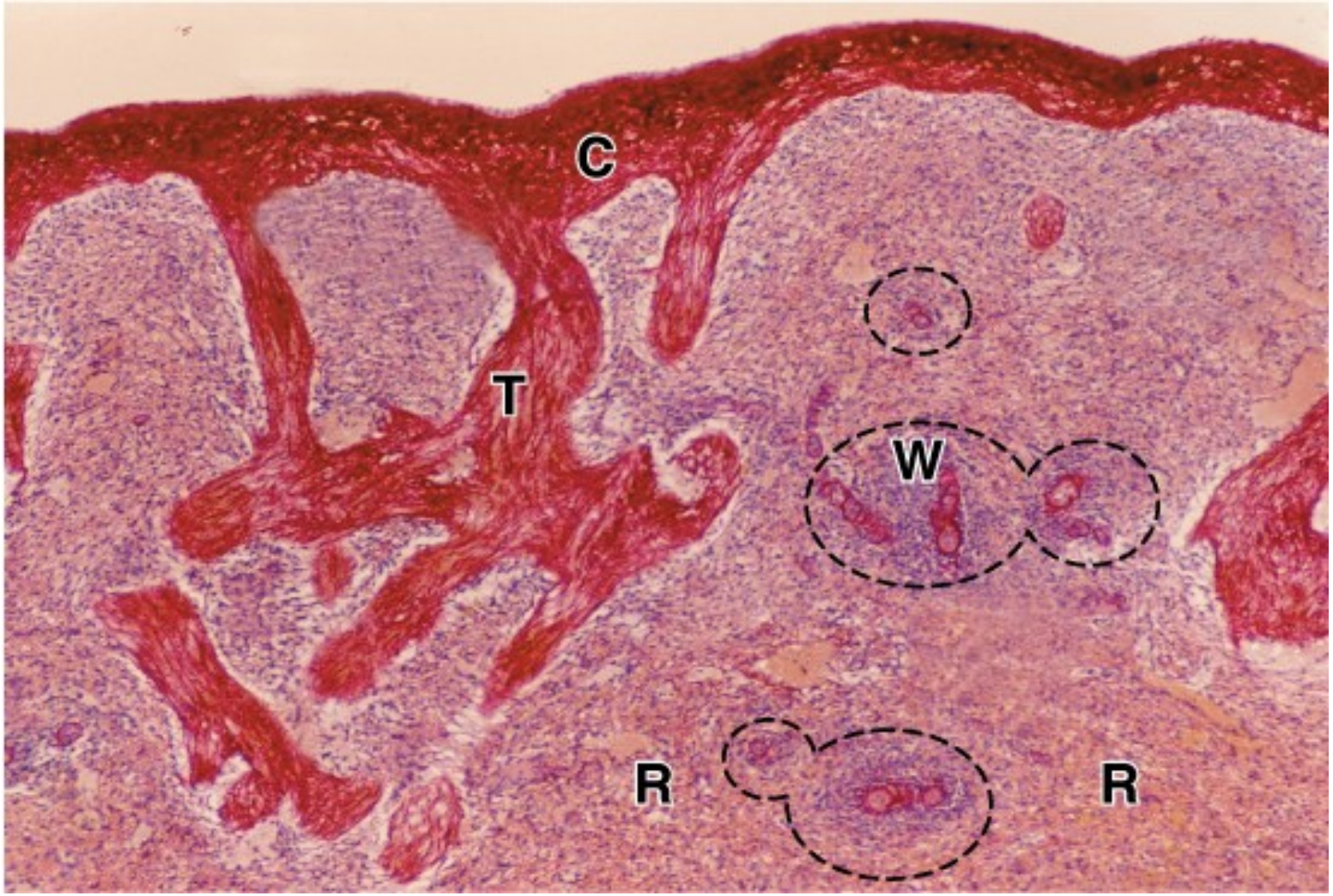
Functions

- It produces lymphocytes & antibodies
- It participate in the body defense system as a site of lymphocyte(white blood cell) proliferation
- **PRENATALLY**= a hematopoietic organ
- **Postnatally**= is involved in identifying, removing + destroying expanded red blood cells+ broken- down platelets, in recycling iron+ globin.
- serves as a blood reservoir
- storing RBC+ platelets+ to a limited degree
- Activation and proliferation of B & T cells
- differentiation of B cells and plasma cells
- secretion of antibodies in white pulp

Histological Features

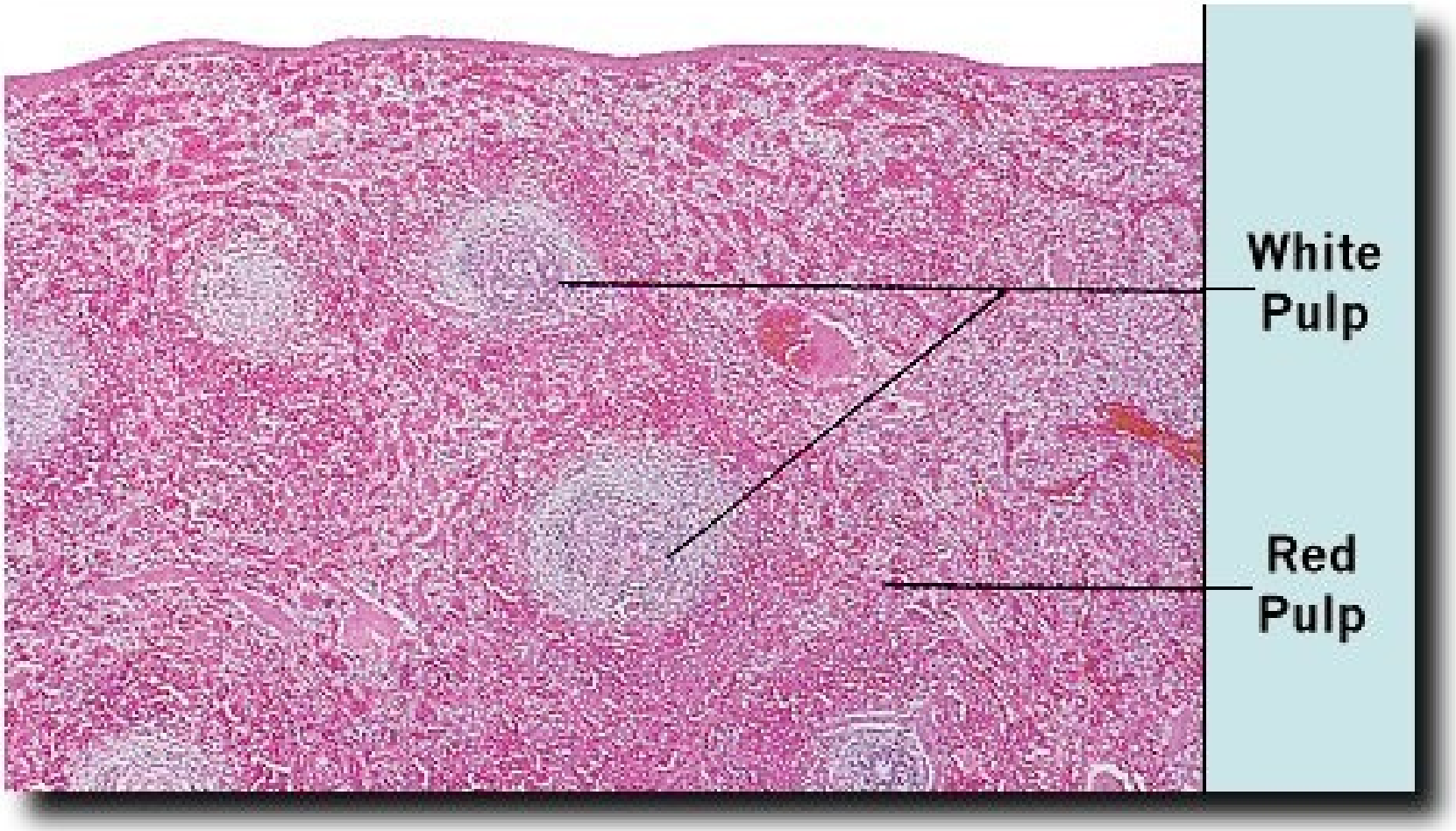
- The spleen is surrounded by a **capsule** of dense connective tissue from which emerge **trabeculae**, which partially subdivide the parenchyma or **splenic pulp** .
- Large trabeculae originate at the hilum, on the medial surface of the spleen.
- These trabeculae carry nerves and arteries into the splenic pulp as well as veins that bring blood back into the circulation.
- Lymphatic vessels that arise in the splenic pulp also leave through the hilum via the trabeculae.

The capsule (C) of the spleen connects to trabeculae (T) which partially subdivide the pulp-like interior of the organ. The red pulp (R) occupies most of the parenchyma, with white pulp (W) restricted to smaller areas, mainly around the central arterioles. Names of these splenic areas refer to their color in the fresh state: red pulp is filled with blood cells of all types, located both in cords and sinuses; white pulp is lymphoid tissue. Large blood vessels and lymphatics enter and leave the spleen at a hilum. X20. PSH.



Splenic pulp

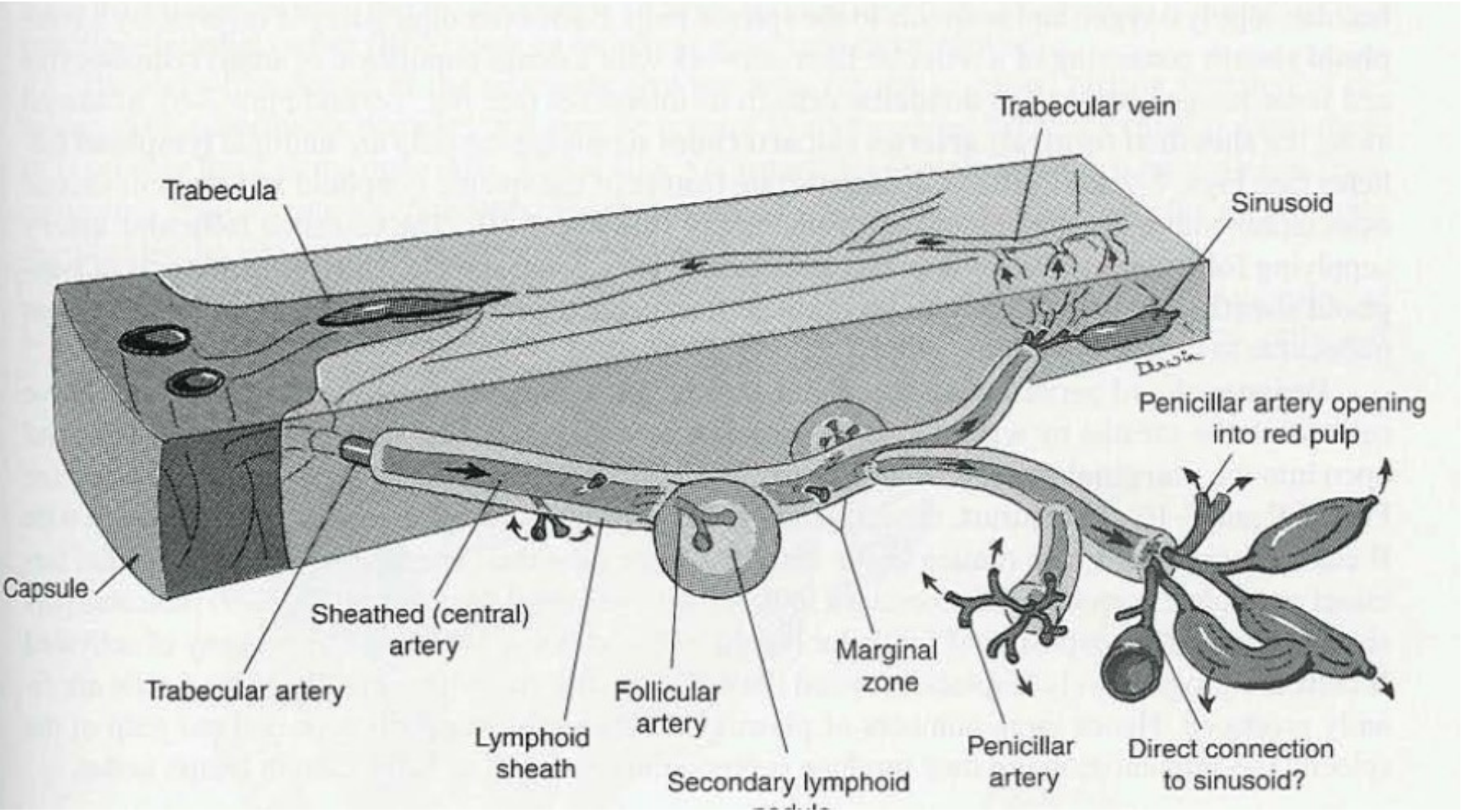
- The spleen is composed of reticular tissue containing reticular cells, many lymphocytes and other blood cells, macrophages, and APCs. The splenic pulp has two components, the **white pulp** and the **red pulp**.
- The small masses of white pulp consist of **lymphoid nodules** and **germinal center**, while the red pulp consists of blood-filled **sinusoids** and **splenic cords** (of Bilroth).

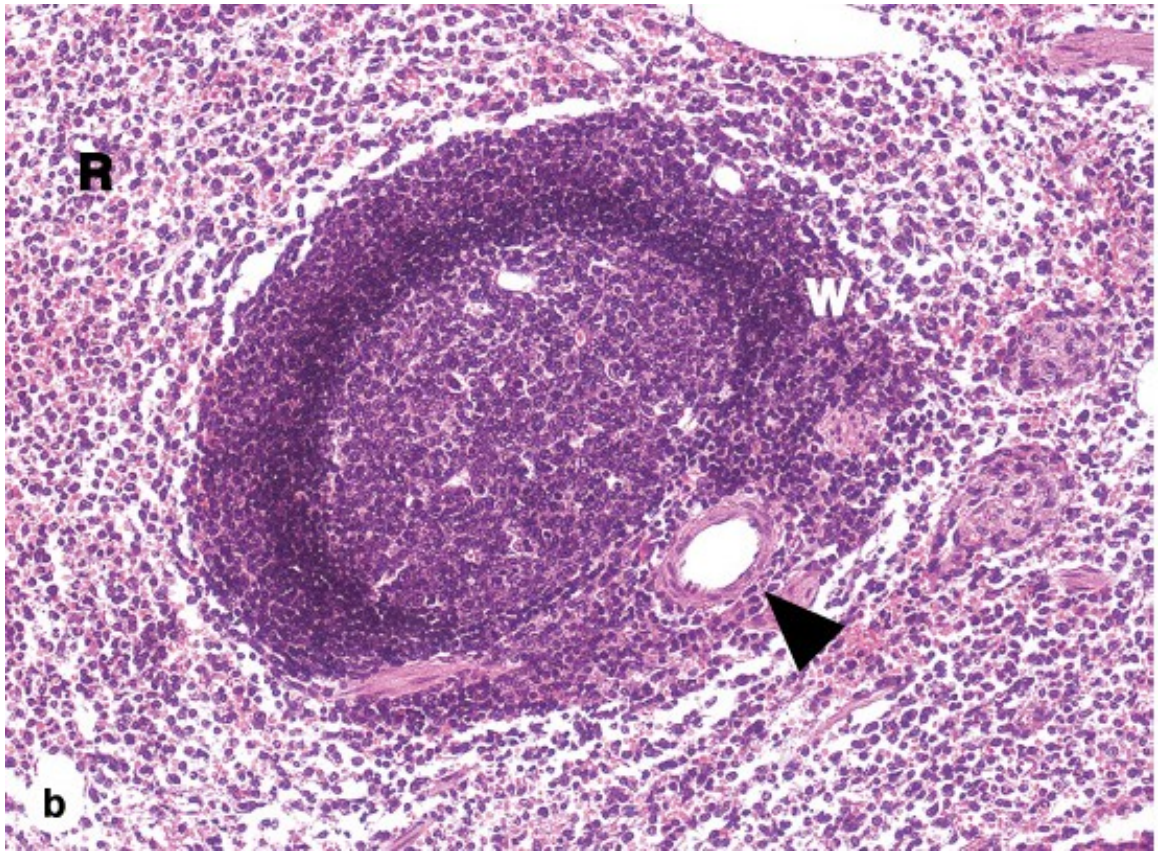

White
Pulp
Red
Pulp

White pulp

- **White Pulp** - lymphatic nodules with germinal center around a central artery.
- The spleen is characterized by numerous aggregation of lymphatic nodules. These nodules constitute the white pulp of the organ.
- The lymphatic nodules also contain, **GERMINAL CENTERS**, that decrease in no. with age.
- The capsule+ trabeculae in the spleen are thicker than those around the lymph nodes+ contain some smooth muscle cells.
- **The large lymphatic nodules constitutes the white pulp of the spleen.** Each nodule normally exhibits a peripheral zone with densely packed small lymphocytes = the ***peri-arterial lymphatic sheath***
- White pulp is surrounded by marginal zone that contains both B & T lymphocytes plasma cells , macrophages and dendritic cells

Diagram showing the association of white pulp with splenic artery

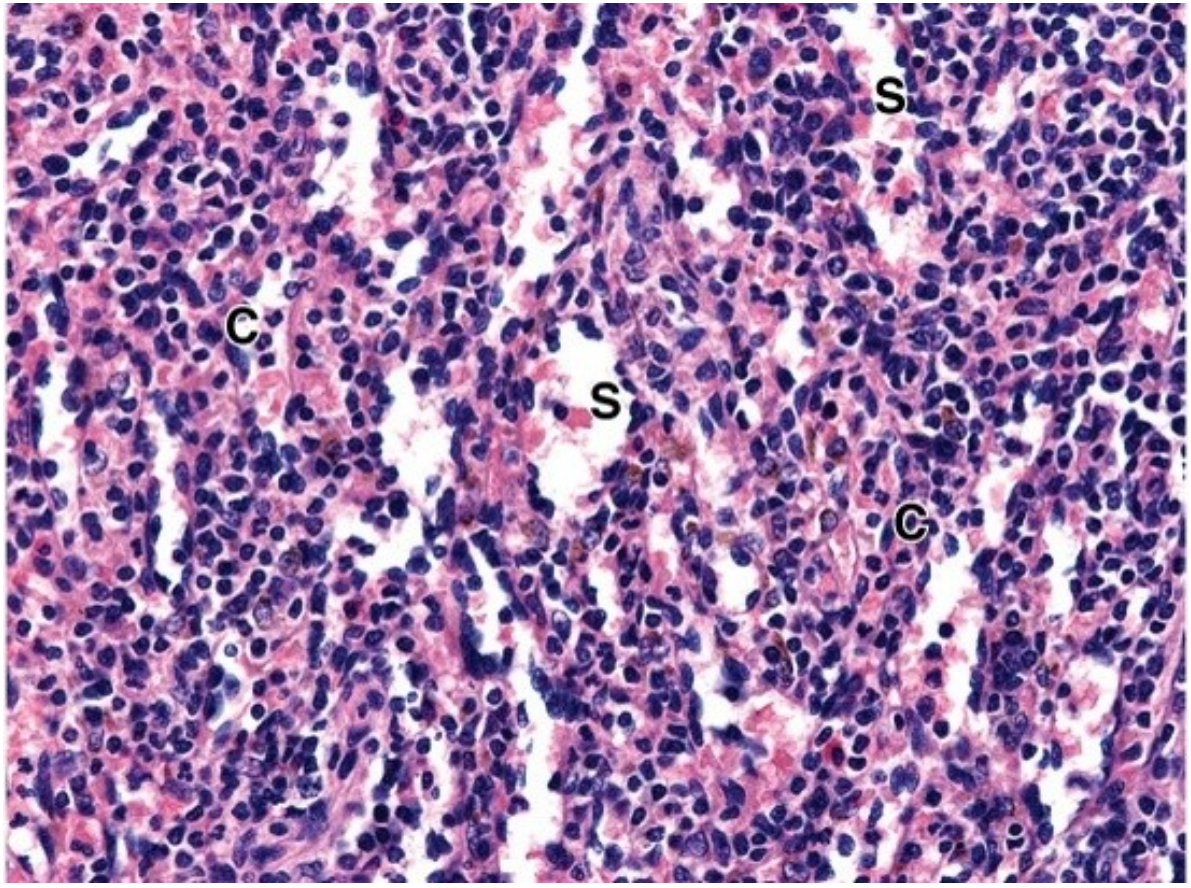




A large nodule with a germinal center forms in the PALS and the central arteriole is displaced to the nodules periphery. Small sinuses can be seen at the margin between white and red pulp.

Red pulp

- Red Pulp – consists of splenic cords+ splenic(blood) sinuses that courses b\w the cords
- **Splenic cords** are thin aggregation of lymphatic tissue containing small lymphocytes, associated cells+ various blood vessels.
- contain monocyte, macrophages, lymphocytes, plasma cells & granulocytes
- **Splenic sinuses** are dilated vessels lined with modified endothelium of elongated cells that appear cuboidal in transverse sections.



The splenic red pulp is composed of splenic venous sinusoids (S) and splenic cords (C), both of which contain blood cells of all types. The cords, often called cords of Bilroth, are reticular tissue rich in lymphocytes. The sinuses are lined by unusual, nonsquamous endothelial cells. X40.H&E. (With permission, from